RESEARCH ARTICLE

Phylogenomics reveals widespread ancient hybridization in the evolution of Lychnophorinae (Vernonieae, Asteraceae)

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Abstract Hybridization is a prevalent evolutionary process in plants and a potential driver of new species formation. Even though hybridization has been studied in many natural systems, the study of genomic signatures of past events of reticulation and hybridization was accelerated by the wide availability of high-throughput sequencing methods. Similar to other species-rich groups of plants, the occurrence of natural hybrids is widely documented in the sunflower family (Asteraceae), yet few studies have focused on ancient hybridization (that occurring between ancestral lineages), mainly due to the lack of comparative genomic data. With the recent increase in phylogenomic studies in Asteraceae made possible by family wide target-capture probes, it has become clear that gene tree incongruence is widespread across the clade, likely reflective of complex speciation events in the past. One clade where natural hybridization is thought to be especially high is the subtribe Lychnophorinae of tribe Vernonieae, with ca. 145 species mostly endemic to the central Brazilian savannas and *campos rupestres*. Here, we for the first time propose a phylogenomic hypothesis for this subtribe and study the impacts of hybridization on its diversification. We newly sequenced 49 species (73 in total) with the Compositae1061 probe set and generated six phylogenetic trees based on different data treatments and using both multispecies pseudocoalescence and concatenated maximum likelihood analysis. To test whether a phylogenetic network, indicative of past and current reticulation events, better fits the data, we used a staggered sampling approach with SNaQ to overcome computational limitations. While overall our analyses confirm relationships recovered in previous analyses, we found five main foci of phylogenetic incongruence and hybridization across the tree. We explore how these reticulation events change our interpretation of some lineages in the subtribe, such as the *Piptolepis pabstii/P. rosmarinifolia* and the *Lychnophora penninervia* complex, further discussing potential taxonomic changes.

Keywords *campos rupestres*; *Eremanthus*; *Piptolepis*; phylogenetic networks; reticulation

Supporting Information may be found online in the Supporting Information section at the end of the article.

■ INTRODUCTION

The role of hybridization in plant speciation and diversification has been long debated (reviewed in Soltis & Soltis, 2009; Folk & al., 2018a; Mitchell & Whitney, 2021; Stull & al., 2023), with differing views on whether reticulation is a diversity-generating process or a force that would lead to evolutionary dead-ends (Wagner, 1970; Schley & al., 2022). Recent hybrids, those involving events among extant taxa, are traditionally identified by indirect evidence such as intermediate morphologies or changes in chromosome number, and have been the focus of botanical studies for centuries, especially in temperate environments, where many species are believed to engage in this process (e.g., Mallet, 2005; Preston & Pearman, 2015).

The steep increase in genomes and phylogenomic datasets in the past couple of decades, brought in by the popularization of second-generation sequencing methods, has revealed that ancient hybridization is widespread in the evolutionary history of plants. However, the deep timescales involved bring challenges to disentangling hybridization from other processes that impact genomic signatures, such as polyploidy, incomplete lineage sorting (ILS), and extinction (Folk & al., 2018a; Hibbins & Hahn, 2022; Stull & al., 2023). In general, ILS can be distinguished from hybridization based on the ratio of genes supporting the inferred species tree topology to those supporting discordant topologies; the process of hybridization creates imbalances in discordant topologies that are unexpected under ILS alone. Approaches to investigate hybridization include simulation, site-based

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methods, and statistical network methods (Folk & al., 2018a; Hibbins & Hahn, 2022). Methods based on the inference of phylogenetic networks (e.g., PhyloNet, SNaQ on PhyloNetworks) are becoming more widely used with phylogenomic datasets, given the availability of hundreds of nuclear loci to test topological incongruences and their increased flexibility in assessing hypotheses of reticulate evolution.

Naturally occurring hybrids are well known in Asteraceae, with foundational works on hybridization carried out with selected members of the family, e.g., studies in *Crepis* L. by Babcock & Stebbins (1938). Examples of natural hybrids and possible hybrid origins of described taxa have been reported in genera from several tribes, such as *Crepis* in Cichorieae (Kilian & al., 2009), *Symphyotrichum* Nees in Astereae (Brouillet & al., 2009), and *Achillea* L. in Anthemideae (Oberprieler & al., 2009). Most studies in natural hybrids in Asteraceae have relied on techniques such as karyology (Brouillet & al., 2009), artificial crossing (Lowrey, 1986), and AFLP fingerprinting (Guo & al., 2005); some have interpreted incongruences between phylogenetic analysis of chloroplast and nuclear markers as evidence of hybrid speciation (Himmelreich & al., 2008). Despite this long history of hybridization studies in Asteraceae, the inference of reticulation events based on phylogenomic data is still incipient. Asteraceae was one of the first plant families to have a dedicated set of probes for target-capture sequencing developed, the Compositae1061 set (Mandel & al., 2014). In the decade since its release, this probe set has fostered many phylogenomic studies that have confirmed or changed our understanding of tribal relationships and biogeography (e.g., Mandel & al., 2019 and the subsequent tribal classification of Susanna & al., 2020). One almost universal result from phylogenomic studies in Asteraceae is the occurrence of incongruences among different nuclear topologies obtained from diverse phylogenetic analysis methods. This issue stems partially from different treatments of potential paralogous genes, the use of different phylogenetic approaches (usually concatenated matrices vs. coalescent analysis), and different levels of missing data, which in these datasets usually is caused by unequal recovery of genes among taxa, creating gappy matrices (reviewed in Herrando-Moraira & al., 2018; Jones & al., 2019).

However, gene vs. species tree conflicts observed in coalescent-based analysis (e.g., Jones & al., 2019; Siniscalchi & al., 2019; Watson & al., 2020; Moore-Pollard & Mandel, 2023) are likely not attributable to methodological issues alone. Several recalcitrant nodes of the Asteraceae tree show high levels of conflict and have proven resistant to differing data, data treatments, and analyses. One example is the phylogenomic analysis of Vernonieae from Siniscalchi & al. (2019) in which the node involving the three South American clades, Lychnophorinae, Chrestinae, and Vernoniinae plus allies, is unresolved in four alternative analyses; a higher number of discordant gene trees were found in these nodes versus the remainder of relationships. Whereas gene/species tree discordance is usually attributed to ILS, it can occur in conjunction with other processes, such as hybridization (Maddison, 1997). In practice, distinguishing between hybridization and ILS in phylogenetic trees is complicated, as they both tend to produce short branch lengths and discordance. If abundant genomic data is available, tests like ABBA-BABA and f_3 and f_4 statistics can be applied to define the roles of each process (reviewed in Kong & Kubatko, 2021). If a high number of gene trees is available (e.g., in transcriptome-based studies), branch length tests such as QuIBL can be implemented (Edelman & al., 2019).

Ancient hybridization, that involving nodes ancestral to extant species, is particularly important to assessing the role of hybridization in diversification as it provides a long-term view on the ultimate evolutionary success of hybrids that is less clear in hybridizing complexes among extant species (Folk & al., 2018a). Despite the increase in phylogenomic datasets of Asteraceae, few studies have investigated the occurrence of ancient hybridization events and have instead focused on groups with known history and evidence of present-time hybridization, leaving us without a needed long-term viewpoint. Watson & al. (2020), in addition to identifying high levels of discordance in a sparsely sampled phylogeny of five tribes (Anthemideae, Astereae, Calenduleae, Gnaphalieae, Senecioneae) with notoriously hard-to-resolve relationships and extensive records of natural hybrids, also identified four potential events of ancient hybridization. In an analysis of the Andean genus *Diplostephium* Kunth (tribe Astereae) with ddRAD sequencing, Vargas & al. (2017) found significant incongruence and strong signals of hybridization. Moore-Pollard & Mandel (2023) investigated sources of phylogenetic conflict in *Packera* Á.Löve & D.Löve (Senecioneae) and found strong signal of hybridization associated with difficult-to-resolve clades. Many similar recalcitrant Asteraceae nodes are candidates for ancient reticulation events in need of further investigation.

Cytogenetics has helped to identify both natural (Faust, 1972) and artificial (Jones, 1976) hybrids in tribe Vernonieae, and hybridization in combination with polyploidy has been invoked as a possible cause for the high variability of chromosome numbers in the tribe (Angulo & al., 2022). The literature shows reports of intermediate morphotypes and putative hybrids in populations of several genera of the tribe, such as *Lychnophora* Mart. and *Eremanthus* Less. (Loeuille & al., 2019), *Elephantopus* L. (James, 1959), *Vernonia* Schreb. (Urbatsch, 1977), and intergeneric hybrids between *Eremanthus* and *Paralychnophora* MacLeish and between *Eremanthus* and *Piptolepis* Sch.Bip. (Loeuille & al., 2015), indicating that hybridization may be a common feature in this group. Despite the cumulative evidence of natural, present-day hybrids in Vernonieae, the occurrence of more ancient hybridization events in the tribe has not yet been tested with genomic data or in phylogenomic studies.

In Vernonieae, the subtribe Lychnophorinae is a hotspot of putative natural hybrids, especially in the genera *Lychnophora* and *Eremanthus*. This subtribe contains 19 genera and ca. 145 species of mostly shrubs and treelets from the *cerrado*

and *campos rupestres* of eastern Brazil (Fig. 1). Syncephaly (i.e., secondary capitula) and the thick indumentum covering leaves and stems make this striking lineage a model for the study of evolution of protection against fire and herbivory (Loeuille & al., 2019, 2024). The phylogenetic relationships between members of Lychnophorinae have been investigated by Loeuille & al. (2015) based on morphological and molecular data (nuclear ITS and ETS, and plastid *ndhF* and *rpl32-trnL*), and defined the boundaries of a monophyletic subtribe. While clades treated as genera received high statistical support in this work, the relationships among them remained partially unresolved. These relationships are partially reflected in the phylogenomic approach of Siniscalchi & al. (2019), with a more restricted sampling. In this later phylogenomic work, the relationships between *Minasia* H.Rob., *Piptolepis*, *Eremanthus* and *Lychnophora* varied between analyses (concatenated maximum likelihood or pseudocoalescence), and missing data treatments. The low support of internal relationships inside Lychnophorinae was interpreted as a result of poor taxonomic sampling and rapid diversification (Siniscalchi & al., 2019). These past studies diverge on the position of *Eremanthus*, for which the sister group is recovered as *Lychnophora* (Loeuille & al., 2015) or *Lychnophorella* Loeuille & al. (Siniscalchi & al., 2019).

The analyses of Loeuille & al. (2015) also recovered incongruences between nuclear and plastid trees, involving the placement of species of *Eremanthus*, *Paralychnophora*, and *Piptolepis*, besides providing evidence for the occurrence of two natural putative intergeneric hybrids, one of them between *Eremanthus erythropappus* (DC.) McLeish and *Piptolepis ericoides* Sch.Bip. (Loeuille & al., 2015). In the same study, *Eremanthus pabstii* G.M.Barroso emerged as sister to *Piptolepis* and was subsequently transferred to the latter based on some morphological characters (pad-like leaf sheath, 3- to 5-armed swollen trichomes, biseriate pappus) despite the presence of other characters at odds with the circumscription of *Piptolepis* (i.e., presence of syncephalium, one floret per capitulum, geographical distribution) (Loeuille & al., 2019). A closely related species was later described (*P. rosmarinifolia* Bringel & al.), showing similar morphological incongruences (Bringel & al., 2019). Marques & al. (2022), while studying the anatomy of cypselae and pappus of *Piptolepis*, noticed that *P. pabstii* (G.M.Barroso) Loeuille & al. and *P. rosmarinifolia* differ from the rest of genus by having cypsela twin hairs with bifurcated apex and no crystals in the collenchymatic mesocarp of the cypsela. Motivated by these results, taxonomic studies of *Piptolepis* were undertaken (Bonfim e Cândido & Loeuille, 2021, 2022) leading to an increase in the number of species, now totaling 19 (Bonfim e Cândido & al., 2025).

With the accumulated evidence of topological incongruences in phylogenetic analyses of Lychnophorinae, the observed evidence of natural hybrids occurring in at least two genera (i.e., *Eremanthus* and *Piptolepis*), and the interesting occurrence of syncephaly in two species of *Piptolepis*, the subtribe is an ideal target for phylogenomic analyses and exploratory analysis about ancient hybridization. For this, a larger sampling of *Piptolepis* and *Eremanthus* is needed to understand relationships among genera and explore the impact of ancient hybridization events on the phylogenetic relationships of Lychnophorinae. The objectives of our study are: (1) to test the monophyly of the genus *Piptolepis* and its relation to *Eremanthus*; (2) to clarify the phylogenetic position of *P. pabstii* and *P. rosmarinifolia*; and (3) to explore the extent of past hybridization events in the evolutionary history of the subtribe Lychnophorinae and its impact in phylogenetic analyses. Our data offer a framework for dissecting complex evolutionary dynamics amidst ancient gene flow, underscoring the need for additional models and tools for phylogenomics.

■ MATERIALS AND METHODS

Sampling. — A total of 73 species were sampled, 69 of which belonging to subtribe Lychnophorinae and representing 18 of the 20 genera (ca. 47% of the species diversity of the group). Of these 69 samples, 49 were newly sequenced in this study, with the remainder obtained from publicly available sequences from Mandel & al. (2019) and Siniscalchi & al. (2019) (NCBI Sequence Read Archive BioProjects PRJNA540287 and PRJNA546287). *Pseudostifftia kingii* H.Rob., belonging to tribe Moquinieae, was selected as the outgroup, and *Vernonanthura polyanthes* (Spreng.) A.J.Vega & Dematt., *Piptocarpha macropoda* (DC.) Baker, and *Allocephalus gamolepis* Bringel, J.N.Nakaj. & H.Rob. were chosen as non-Lychnophorinae Vernonieae terminals. The sampling was focused on optimizing genus and species representation, not on population-level sampling, thus only one specimen per taxon was included. Voucher details are available in Appendix 1.

DNA extraction, library preparation and sequencing. — DNA was extracted from silica-gel dried leaves or from leaves sampled from herbarium materials. DNA was extracted either with the E.Z.N.A. SQ Plant DNA Kit from Omega Biotek (Norcross, Georgia, U.S.A.), with addition of polyvinylpyrrolidone and ascorbic acid to the first extraction buffer (10 ml SQ1 buffer, 100 mg polyvinylpyrrolidone, 90 mg ascorbic acid) or with a CTAB protocol, based on Doyle & Doyle (1987). DNA samples were quantified with fluorimetry using Qubit (Termo Fisher Scientific, Waltham, Massachusetts, U.S.A.) and the need for further fragmentation was evaluated through running the extracted DNA on a 1% agarose gel. Fragmentation was carried out in a QSonica Q500 sonicator (Qsonica, Newtown, Connecticut, U.S.A.) for 5 to 7 minutes, aiming for a 400–500 bp insert size. Libraries were prepared with the NEBNext Ultra II DNA Library Prep Kit for Illumina (New England Biolabs, Ipswich, Massachusetts, U.S.A.) with an initial amount of at least 500 ng of total DNA, using 15 cycles in the last amplification step. A target-capture approach was used, with the Compositae1061 MyBaits probe set (formerly CompCOS, Daicel Arbor Biosciences, Ann Arbor, Michigan, U.S.A.). Libraries were

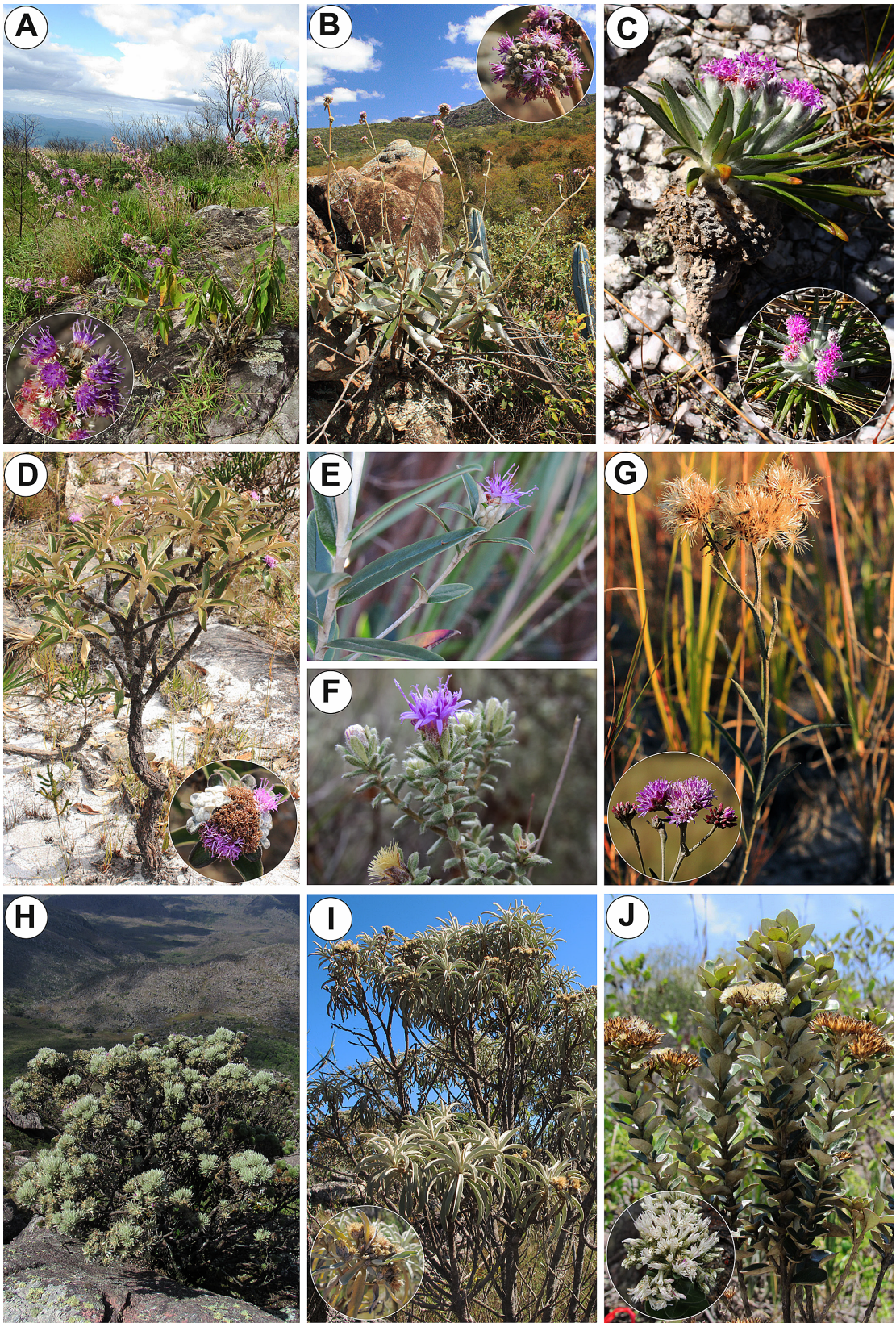

Fig. 1. Diversity of Lychnophorinae, with inflorescence morphology highlighted. **A**, *Cololobus rupestris*; **B**, *Maschalostachys mellosilvae*; **C**, *Prestelia eriopus*; **D**, *Lychnocephalus tomentosus*; **E**, *Lychnophora albertinioides*; **F**, *Piptolepis leptospermoides*; **G**, *Piptolepis pabstii*; **H**, *Lychnophorella sericea*; **I**, *Eremanthus veadeiroensis*; **J**, *Eremanthus hatschbachii*. — Photo credits: A by P. Gonella; B, C, D & G main by C.M. Siniscalchi; E, F, H & J main by J.B. Candido; G detail by J. Bringel; I by B. Loeuille; J detail by H. Ogasawara.

equimolarly pooled in groups of up to 20, with a total amount of DNA of 500 ng. An incubation time of 36 h was used, and the last amplification step used 15 cycles. Quality was checked using Qubit and Bioanalyzer, and sequencing was conducted using Illumina Hi-Seq 2500 in paired-end, high-throughput mode, at Novogene (Sacramento, California, U.S.A.).

Sequence assembly. — De-multiplexed sequences were trimmed with Trimmomatic v.0.33 (Bolger & al., 2014) to remove adaptors and low-quality sequences. The PHYLUCE pipeline (Faircloth, 2016) was used to map reads back to the original loci in the probe set. Sequencing reads were first assembled into contigs using SPAdes v.3.14.1 (Bankevich & al., 2012), with default parameters and kmer lengths of 21, 33, 55, 77, 99, and 127. The contigs were then matched back to the original probe sequences using PHYLUCE and aligned with MAFFT v.7.490 (Katoh & al., 2002) within the pipeline. The final products were aligned individual gene matrices and an aligned concatenated matrix with all genes. PHYLUCE assembly results in long matrices compared to other assembly approaches for target-capture (e.g., HybPiper), as the initially assembled contigs are not trimmed to match the length of the original probe region and the regions flanking the original probes are included. Additionally, PHYLUCE takes a conservative approach in dealing with potentially paralogous genes (Faircloth, 2016) (in this context, loci for which the assembled contigs match multiple loci or different contigs match the same locus from the final alignments) by eliminating them from the assembly on a sample-by-sample case. Therefore, the resulting gene matrices are usually gappy, as the recovery of flanking regions is not equal among samples and genes, and taxon occupancy varies by gene. We additionally ran a HybPiper exon assembly (Johnson & al., 2016), which generated much shorter matrices. A comparison of results of both pipelines is available in suppl. Appendix S1. Alignment summaries were calculated using AMAS (Borowiec, 2016) and are summarized in Table 1 and are fully available on suppl. Table S1.

Phylogenetic analyses. — Phylogenetic analyses were run using two different approaches: concatenated maximum likelihood analysis in RAxML-NG v.0.9.0 (Kozlov & al., 2019) and a pseudocoalescence approach using ASTRAL-III v.5.7.3 (Zhang & al., 2018) with gene trees generated with maximum likelihood. In the concatenated analysis, we first trimmed sites with more than 50% of invariable characters using the "psclsq" function in the package phyx (Brown & al., 2017), then used the function --parse in RAxML-NG to verify the best model for the alignment. The analysis was then ran using the --all option, with bootstraps calculated until convergence. Support values were calculated as bootstrap support (BS). Two analyses were run, one with the full set of genes recovered (hereby called "total"), and one with a smaller number of genes that had a taxon occupancy of at least 68% (hereby "68p"), a sampling level chosen based on the assembly in order to retain at least 50 taxa per gene, to verify the impact of removing genes with high levels of missing data.

Gene trees were obtained by running individual maximum likelihood analysis for each gene in RAxML v.8.0.0 (Stamatakis, 2014) for 100 bootstraps, using the GTR + Γ model for all alignments. Besides analyses using the total and 68p datasets to verify the impact of low-support clades on the coalescence analysis, we ran another set of total and 68p ASTRAL analyses in which the gene trees had clades with bootstrap support lower than 10 collapsed before the analysis to verify the impact on support values (called bs10 in the Results and Discussion). Branch support in ASTRAL analysis was calculated as local posterior probabilities (LPP). All trees are shown rooted with *Pseudostifftia kingii*.

Phylogenetic discordance. — To visualize the topological differences between the six trees, we calculated the Robinson-Foulds distance between each pair of rooted trees on PAUP* v.4.0a (Swofford, 2003). The values were organized in a symmetric matrix, and the multidimensional scaling among values was calculated using the "cmdscale" function in R (v.4.3.1) in RStudio (build 2023.09 Desert Sunflower), and the results were plotted using base R functions. We also used the "densityTree" function of the R package phytools v.1.9-16 (Revell, 2023) to overlap all trees and show the topological differences among them and produced tanglegrams of tree pairs using several functions in this same package. Scripts are available on https://github.com/carol-siniscalchi/lychnophorinae-reticulation.

Phylogenetic discordance between gene trees and the final species tree was formally calculated using the multi-pronged approach of QuartetSampling (QS) (Pease & al., 2018). This quartet-based method provides three metrics at each node that allow the distinction of discordance due to (1) low information, (2) lineage sorting/introgression and (3) rogue taxa. These metrics are namely quartet concordance (QC), which explains how often a concordant quartet is inferred over the two discordant quartets, varying from −1 to 1, quartet differential (QD) that shows if the frequencies of either discordant quartet are skewed, varying from 0 to 1, and quartet informativeness (QI), that shows what proportion of replicates were informative and also varies from 0 to 1. Additionally, quartet fidelity (QF) is informed for the tips, as a measure of how often that terminal taxon produces a

Table 1. Alignment statistics of different PHYLUCE assemblies.

Treatment	PHYLUCE (total)	PHYLUCE (68p)
No. aligned genes	517	114
No. taxa [median (min–max)]	10 (3–68)	58 (50–68)
Alignment length [mean (min–max)]	785.8 (110–6566)	1168.6 (467–6566)
Missing data percentage [mean (min–max)]	22.4 (0–98.78)	36.6 (9.9–89.8)
No. parsimony-informative sites [median (min–max)]	31 (0–301)	91.5 (32–301)

concordant topology when sampled; this measure also ranges from 0 to 1. For this analysis, we chose to focus on the ASTRAL 68p with collapsed low-support clades tree (ASTRAL 68p bs10), with the corresponding concatenated 68p alignment. The QS analysis was carried out with 100 replicates and an unpartitioned alignment.

Hybridization analyses. — To test the occurrence of past introgression events across the evolutionary history of Lychnophorinae, we conducted a hybridization analysis using the SNaQ algorithm in the package PhyloNetworks v.0.16 (Solís-Lemus & Ané, 2016; Solís-Lemus & al., 2017). Due to computational limitations in the number of taxa that can be included in a single PhyloNetworks analysis, we subdivided the tree into seven subtrees, so we could verify the occurrence of past introgression in finer detail. These subsets were defined to investigate taxa that changed positions in different phylogenetic analyses (e.g., *Lychnophora penninervia*) and to better investigate clades such as *Eremanthus* and *Piptolepis*. A total of 47 species were included and tested among these subsets. We obtained seven subtrees containing 12 or 13 taxa by re-running the PHYLUCE pipeline from the "PHYLUCE_assembly_get_match_counts" step, selecting different taxon groups. As described above, gene trees were then obtained with RAxML with 100 bootstraps and species trees were assembled with ASTRAL.

In PhyloNetworks, the concordance factors for each subtree were obtained from the unrooted gene trees using the "countquartetsintrees" function. Each optimization was run 10 times, and the number of hybridization events in each analysis was set from 0 to 8. The number of hybridizations with the lowest negative log-likelihood (best H) was selected as the best one for each subtree and inheritance probabilities for each event were calculated (shown in Table 2). The number of events recovered in the tree does not always match the best log-likelihood number of events (e.g., analysis 3 "Eremanthus", with a best H of 8 events recovers only 4 events in the tree). Bootstrap analysis for the network with the best number of events was carried out with the bootstrap trees obtained from RAxML. For each analysis, we ran the TICR test (The Incongruence Checking in R; Stenz & al., 2015) as implemented in the julia package QuartetNetworkGoodnessFit v.1.0.0 (Cai & Ané, 2021). This test measures how well a tree model including ILS, such as the one generated by ASTRAL, explains the gene discordance. The test takes as inputs the concordance factors calculated previously and the species tree without the calculated reticulations. In the context of this analysis, bootstrap values represent the percentage of bootstrap trees showing the same sister-hybrid relationship as the reference network, and it is represented by two values, one for the hybrid edge and one for the major sister edge. For the purposes of this study, we follow the definition of ancient hybridization/reticulation given in Folk & al. (2018a); namely as those hybridization/reticulation events that involve deep nodes of the trees. Thus, recent hybridization/reticulation under this framework involves tips of the tree (i.e., species). The classification of an event as recent does not necessarily imply that populations of these species are actively hybridizing in the present, only that hybridization postdated past species divergence events.

■ RESULTS

The sequencing of 49 new samples of Lychnophorinae generated between 503,369 and 19,009,661 reads per sample. The PHYLUCE pipeline generated a total of 517 recovered genes (the total dataset), and the 68p dataset contained 104 genes. Alignments varied from 110 to 6566 base pairs, and the percentage of parsimony-informative sites varied from 0 to 28.9 (suppl. Table S1).

All six topologies produced by the different treatments and phylogenetic approaches here have slight differences among them, which are concentrated in a few clades (suppl. Figs. S1–S6). Regardless of phylogenetic analysis method, the three largest genera in the subtribe, *Lychnophora*, *Eremanthus*, and *Piptolepis*, are recovered as non-monophyletic. By

Table 2. Log-likelihood values recorded for each number of hybridization events in each PhyloNetworks analysis.

	Analysis						
H number	Basal	Cauliflorus	Eremanthus	Lychnocephalus	Mixed	Penninervia	Piptolepis
H0	797.8449	2210.0572	1145.0198	1526.9009	1716.1775	926.3458	1831.0396
H1	688.5765	1946.0501	663.7656	1394.0220	1498.6177	580.2002	1673.9591
H2	666.6909	1811.3861	642.6213	1359.9935	**1492.0364**	530.9619	**1670.1674**
H3	618.5401	**1768.3004**	626.7223	**1315.4705**	1492.0354	530.9619	1669.9260
H4	**592.5718**	1767.5944	624.1141	1315.4705	1492.0354	**523.2747**	1669.7701
H5	592.5387	1767.5233	612.1468	1315.4705	1492.0354	0	1669.7150
H6	592.5052	1767.5216	0	1315.4705	1492.0354	533.9366	1669.7080
H7	592.4678	1767.5192	0	1315.4705	1492.0354	532.1175	1669.4753
H8	592.4347	1767.5109	**609.4501**	1315.4705	1492.0354	532.1152	1593.2695

Values in bold represent the chosen number of hybridization events.

overlapping the six topologies, it is possible to see that the main foci of differences are (1) within the *Piptolepis* clade, (2) the relative position of *Eremanthus elaeagnus* (Mart. ex DC.) Sch.Bip. within the larger clade formed by the bulk of *Eremanthus*, *Lychnophora* and *Lychnophorella*, and (3) the position of a clade formed by *Hololepis pedunculata* (DC. ex Pers.) DC., *Anteremanthus piranii* Roque & F.A.Santana and *Eremanthus crotonoides* (DC.) Sch.Bip. in relation to the remaining clades (Fig. 2A). The multidimensional scaling of Robinson-Foulds distances shows that among the four ASTRAL trees, the total datasets are more similar to each other and the 68p datasets are more similar to each other, than total is to 68p. The concatenated analyses, on the other hand, are significantly different from the ASTRAL topologies and from each other on the MDS2 axis (Fig. 2B).

Despite incongruences in more nested relationships, the backbone of the tree is mostly stable (Fig. 2A). *Cololobus rupestris* (Gardner) H.Rob., a morphologically unusual species from the Atlantic Forest, with slightly succulent leaves and corollas with very short lobes, emerges successively from *Centratherum* Cass. and is sister to the rest of the subtribe. *Anteremanthus piranii* emerges in a clade with *Hololepis pedunculata* and *Eremanthus crotonoides*, while *Anteremanthus hatschbachii* H.Rob. emerges in a more nested clade containing *Paralychnophora* and *Maschalostachys* Loeuille & Roque. The two species of *Heterocoma* DC. sampled here always form a clade with *Chronopappus bifrons* (DC. ex Pers.) DC. *Proteopsis argentea* Mart. & Zucc. ex Sch.Bip. and a monophyletic *Minasia* emerge successively as sister to the bulk of the subtribe, which is divided into five clades. One clade shows *Prestelia eriopus* Sch.Bip. as sister to a clade formed by a few *Lychnophora* species plus three species of *Lychnocephalus* Mart. ex DC. This clade is reciprocal sister to the clade containing all but two species of *Piptolepis*. The last large clade is sister to this *Lychnocephalus* + *Piptolepis* clade. Within it, most of the sampled *Lychnophora* form a clade that is sister to the rest. Finally, all sampled *Lychnophorella* form a clade with a couple of *Eremanthus*, while another clade contains seven species of *Eremanthus* plus *Piptolepis pabstii* and *Piptolepis rosmarinifolia*.

Among the four ASTRAL topologies, the two obtained with the 68p dataset present fewer conflicts between them than the total dataset topologies (suppl. Figs. S7, S8). Conflicts in the infrageneric relationships in *Piptolepis* are abundant between all topologies and the nodes in this clade have generally low support values (Fig. 3B); however, in all trees, the *Piptolepis* clade is divided into three subclades. In the ASTRAL topologies, *Eremanthus elaeagnus* appears either as sister to a clade formed by *E. erythropappus*, *E. capitatus* (Spreng.) MacLeish, and *E. uniflorus* MacLeish & H.Schumach., or as sister to a clade formed by *Lychnophora penninervia* Sch. Bip., *E. incanus* (Less.) Less., and *E. auriculatus* MacLeish & H.Schumach., plus two species of *Piptolepis*. On the other hand, in the concatenated trees, *L. penninervia* is the "floater" species (suppl. Fig. S9). Comparing the ASTRAL 68p with collapsed low-support clades topology (suppl. Fig. S2) with the total concatenated tree (suppl. Fig. S6), three main differences emerge: (1) in the concatenated analyses, *Anteremanthus hatschbachii*, instead of grouping in the *Paralychnophora* clade, emerges together with *A. piranii* in a clade with *Hololepis pedunculata* and *Eremanthus crotonoides*, (2) *Lychnophora penninervia* emerges in the *Lychnophora* clade in the concatenated analysis, and (3) in all ASTRAL analysis, *Allocephalus gamolepis* Bringel & al., which is placed in subtribe Dipterocypselinae, emerges outside of Lychnophorinae and sister to all sampled Vernonieae, while in the concatenated analysis it appears more internally as sister to all Lychnophorinae.

The Quartet Sampling analysis (Fig. 3A) further emphasizes these points of topological conflict among different analyses; while most of the nodes of the tree show strong quartet concordance (most of all quartets support the focal branch), the nodes involved in topological disagreement show weak support with a discordant skew (low support of quartets for the relationship shown with a supported alternative topology) or show counter-support (a majority of quartets support one of the alternative discordant topologies). Of the 14 internal nodes in the *Piptolepis* clade, four show counter-support (in the form of negative QC values; nodes in yellow in Fig. 3A) and one shows weak support with a discordant skew (defined by low QC and high QD value; node in turquoise). The *Lychnophora* and *Eremanthus* clades also show four nodes with weak or counter-support each (nodes shown in yellow and turquoise in Fig. 3A). In *Eremanthus*, the uncertainty is concentrated on the clade containing *L. penninervia*, *E. incanus* and *E. auriculatus*. Only one node in the backbone shows counter-support, the split between *Proteopsis argentea* and the rest of the subtribe, an interesting fact, as this taxon emerges in this same position in five of the six trees, although with varying support (0.6 to 0.95 LPP in ASTRAL trees and 77% BS in the total concatenated analysis). The QF values, which indicate how often a certain taxon produces a concordant topology, vary from 0.43 to 0.86 across the tree. The lowest values, which indicate that the taxon produces more discordant topologies, are associated with the low QC nodes in *Piptolepis*, *Lychnophora* and *Eremanthus*. *Proteopsis argentea*, despite the apparent topological stability, has the lowest QF value, indicating that it produces concordant topologies in 43% of the quartets where it is sampled.

The SNaQ analyses implemented in PhyloNetworks show extensive occurrence of reticulation events in the evolutionary history of Lychnophorinae (Fig. 4). Each of the SNaQ analyses indicated two to four events in each subtree (Table 3). When counted together, they total 20 unique events, with 8 being recent (involving taxon edges) and 12 ancient events (involving only clade edges or clade and taxon edges). One recent event was duplicated in two analyses (*Lychnophora ericoides* Mart. vs. *L. penninervia*), with maximal bootstrap support in one of them (Fig. 4C). Support differed in placing the remaining hybridization edges, some of which had significant uncertainty (for example, Fig. 4F) even if evidence for the presence of hybridization was decisive. The *Piptolepis*

clade is involved in six events, five ancient and one recent. Among them, there is one event involving the *P. redacta* J.B.Candido & Loeuille lineage and the clade formed by *P. pabstii* and *P. rosmarinifolia*, which is deeply nested within *Eremanthus* in the species tree, although with null bootstrap support for this hybrid edge, indicating significant uncertainty. *Piptolepis campestris* Semir & Loeuille and *P. pseudomyrtus* (A.St.-Hil.) Sch.Bip. appear as hybrid edges in one event each (Fig. 4D,E). Four events have their hybrid edges within the *Eremanthus* clade, two of them involving the nested *P. pabstii* and *P. rosmarinifolia* clade. *Proteopsis argentea* is identified as the hybrid edge in one recent event, with *Chronopappus bifrons* as the parent edge (Fig. 4B). It is additionally involved in an ancient event in which the

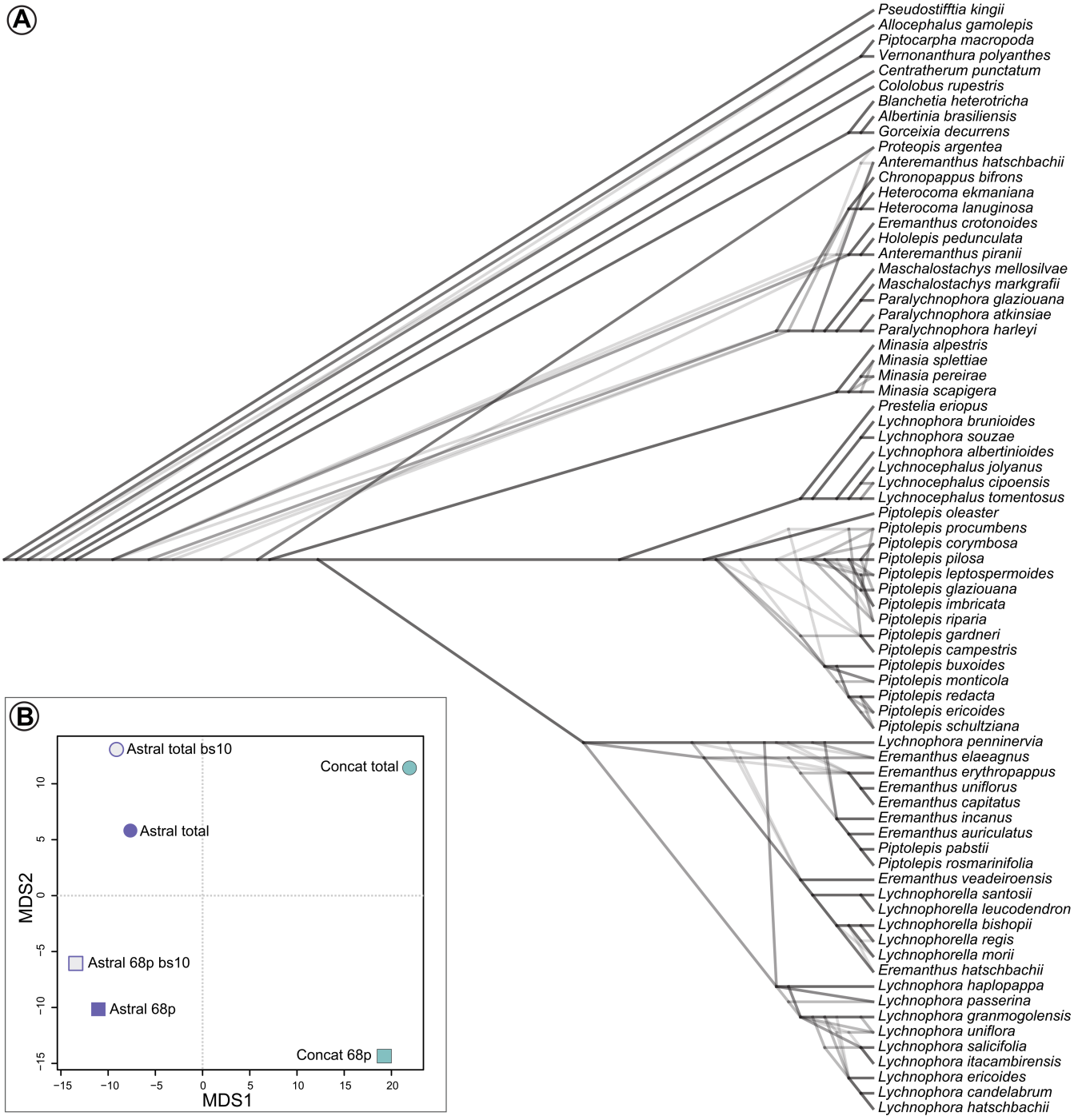

Fig. 2. Topological incongruence among six different phylogenetic analyses of Lychnophorinae, using ASTRAL and concatenated maximum likelihood. **A**, Overlap of the six topologies, different transparencies of the branches show where topologies agree and disagree; **B**, Results of multidimensional scaling of Robinson-Foulds distances among the six trees.

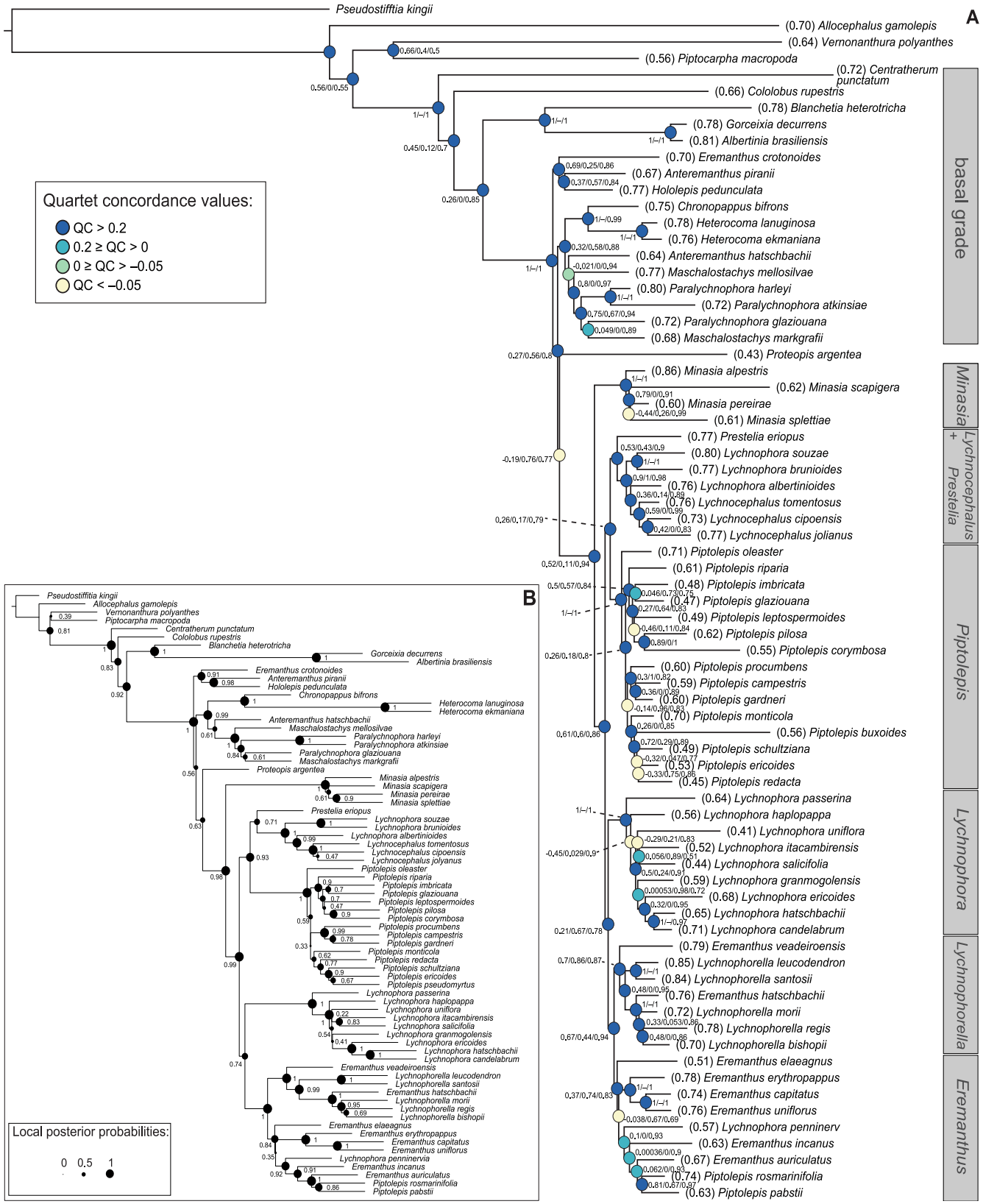

Fig. 3. A, Phylogenetic discordance estimated with Quartet Sampling and the ASTRAL 68p bs10 topology. Scores are shown on branches indicating quartet concordance/quartet differential/quartet informativeness. Quartet concordance is color-coded according to the legend, intervals follow Pease & al. (2018). Quartet fidelity scores are shown in parentheses before the taxon names on the tips. **B**, ASTRAL 68p bs10 topology with local posterior probability support values shown on nodes.

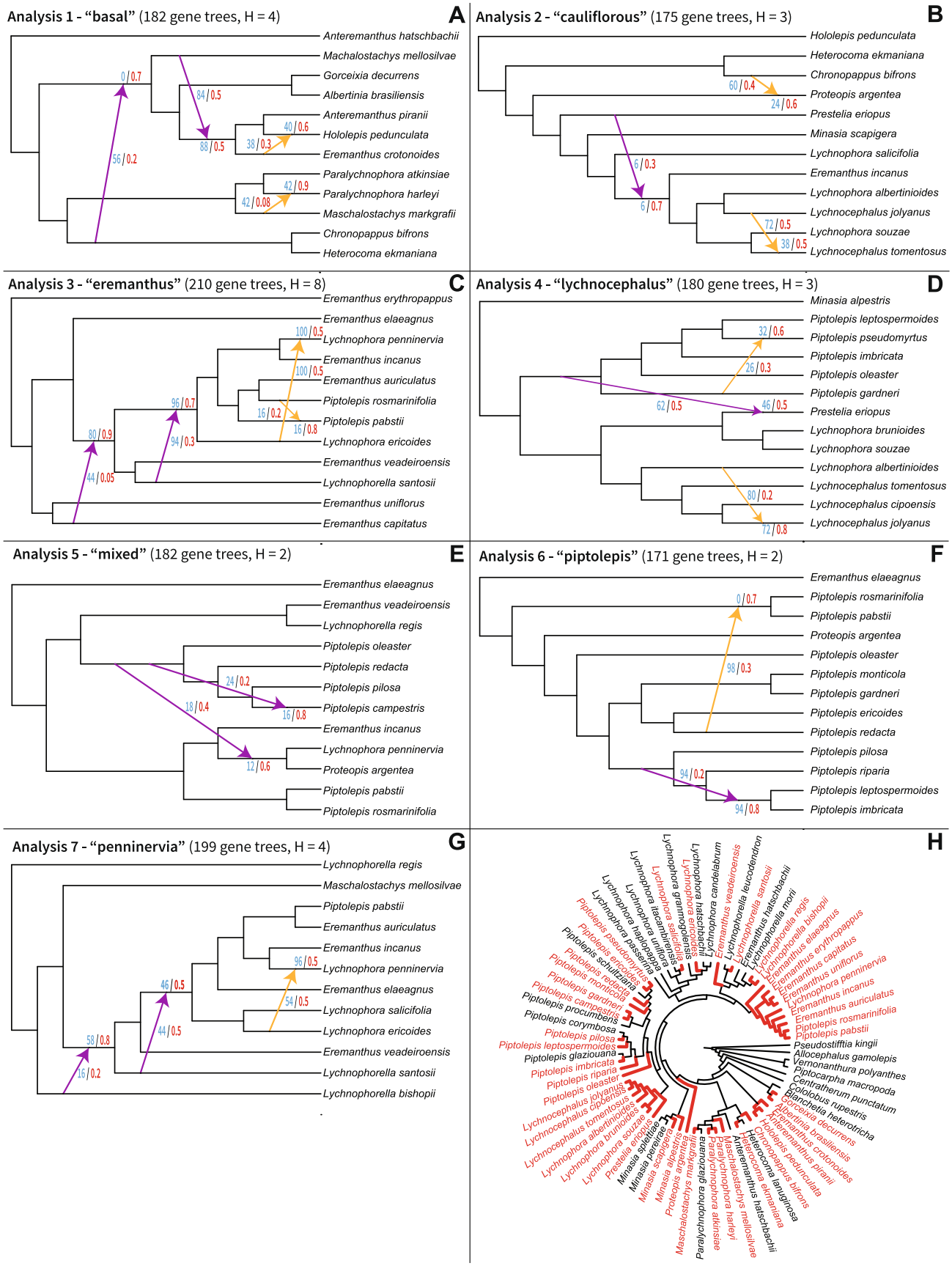

Fig. 4. Caption on next page.

species emerge within the *Eremanthus* clade, thus representing a complicated scenario not reflected in the full species tree. The *Lychnocephalus* + *Prestelia* clade is involved in four events (two recent, two ancient), three of them within members of the clade and one involving the stem of the *Piptolepis* clade. The TICR test results indicate that among the seven analyses, only one ("mixed") could be correctly interpreted as a bifurcating species tree, instead of a network. This analysis differs from the other from showing only ancient reticulation events, both with very low bootstrap values, indicating that incongruences in these gene trees are most likely caused by ILS.

■ DISCUSSION

The impact of ancient hybridization on the evolutionary history of Lychnophorinae. — Our study is the first to test the occurrence of ancient hybridization in a group of Asteraceae mostly endemic to Brazil and the *campos rupestres*. While high degrees of incongruence are not uncommon in phylogenomic investigations in Asteraceae (e.g., Watson & al., 2020; Siniscalchi & al., 2023) and low statistical support persists even when employing hundreds of genes (Thapa & al., 2020), the extensive occurrence of reticulation recovered here stands out. The internal relationships of Lychnophorinae proved surprisingly hard to resolve with a phylogenomic dataset. While the general outline of previously recognized clades was recovered, the relationships among these clades differ from those found in a previous study of molecular and morphological phylogenies (Loeuille & al., 2015). We find that ancient reticulation is a key process behind the recalcitrant relationships in this group and identify several hotspots of Lychnophorinae where low clade support, incongruence, and ancient hybridization overlap (Figs. 2–4). We can define five main focal points of conflict and hybridization: the early-divergent Lychnophorinae, the *Lychnocephalus* + *Prestelia* clade, the *Piptolepis* clade, the *Lychnophorella* and allies clade, and the *Eremanthus* clade.

The early-divergent Lychnophorinae, here designated as a series of clades containing members of *Anteremanthus*, *Heterocoma*, *Maschalostachys*, *Paralychnophora* and a couple of monotypic genera (named "basal grade" in Fig. 3), presents several nodes with low to medium support, a couple of nodes with medium to high conflict, and five reticulation events, mostly among members of this informal group (Fig. 4A,B). Two of these events are recent, and potentially indicate the occurrence of introgression between *Eremanthus crotonoides* and *Hololepis pedunculata*, and *Maschalostachys markgrafii* and *Paralychnophora harleyi*. However, most events in this basal grade have low bootstrap support. The *Lychnocephalus* + *Prestelia* clade and *Piptolepis* clade emerge as reciprocal sisters and concentrate the highest number of reticulation events: nine. From these, only two involve taxa from outside this larger clade, one possibly pointing to an ancient event at the ancestors of the core Lychnophorinae clade (suppl. Fig. S10) and one with the *Piptolepis rosmarinifolia* and *P. pabstii* lineage as the hybrid edge (Fig. 4F). Similarly, the *Lychnophorella* and *Eremanthus* clades are also reciprocal sisters, and from the eight reticulation events involving these lineages, four potentially involve ancient events at the backbone of the tree. Notably, the *Eremanthus* clade is the recipient of four hybrid edges. These reticulation hotspots, not detected previously, would benefit from densely sampled populational-level studies to better delimit species and genus boundaries.

Six of the reticulation events have their hybrid edges in the stem of clades that do not correspond to clades found in the species tree. The main reason for the recovery of these

Table 3. Information on the number of taxa, number of starting gene trees, number of hybridizations (H) with the best log-likelihood, number of reticulation events identified in each SNaQ analysis, and the TICR *p*-value of the respective analysis.

Analysis	No. taxa	No. gene trees	Chosen H value	No. events	TICR *p*-value
Basal	12	182	4	4	0.78034
Cauliflorus	12	175	3	3	0.09871
Eremanthus	12	210	8	4	0.91805
Lychnocephalus	13	180	3	3	0.10674
Mixed	12	182	2	2	0.00314
Piptolepis	13	171	2	2	0.96745
Penninervia	12	199	4	3	0.96442

Fig. 4. A–G, Evidence of hybridization events in Lychnophorinae obtained from multiple analyses on PhyloNetworks. Arrows indicate the direction of the event, with the hybrid edge indicated by the arrowhead. Purple arrows indicate ancient events (between two clade edges or a clade edge and a taxon edge) and orange arrows represent recent events (between two taxon edges). Bootstrap values are shown in light blue and inheritance probability values are shown in red. **H**, ASTRAL 68p bs10 topology with species used across PhyloNetworks analysis highlighted in red.

mismatched clades in the PhyloNetworks analysis is the fact that a new "best network" is calculated in each analysis based on the gene trees fed into SNaQ. Because in each analysis we are only including a small portion of the whole sampling of the tree, it is expected that some relationships will not match perfectly with those found in the species tree. Additionally, the presence of potential reticulations can interfere with coalescent-based methods like ASTRAL (Solís-Lemus & al., 2016), a factor that could also separately contribute to the variation found among different topologies. Because pseudo-coalescence methods like ASTRAL assume no introgression, datasets with high levels of incongruence, like ours, can lead to failure in solving relationships. As shown by the goodness-of-fit TICR test, among the seven analyses, ILS seems to be the dominant cause of incongruence in only one of them. Reticulation events that flow from internal branches to outer edges of the tree can potentially indicate the occurrence of reticulation events in an unsampled or extinct lineage (such as the event involving *Prestelia eriopus* in Fig. 4D). In suppl. Fig. 10, we show all reticulation events plotted over the full species tree, and we point these six events to the stem of the clade that contains all taxa involved in that event, to highlight this possibility.

Considering incongruences caused by different treatments of missing data and analysis method, the multidimensional scaling of Robinson-Foulds distances (Fig. 2B) shows that topologies obtained with the two different datasets (total and 68p) tend to group by dataset, showing stronger similarity: the total topologies separate on the positive pole and the 68p on the negative pole. Collapsing the low-support branches in genes trees affects the 68p less than the total dataset, which informed the decision to select the ASTRAL 68p bs10 topology to further analyze phylogenetic incongruence and hybridization. The large distance between the concatenated maximum likelihood topologies shows that the ASTRAL multispecies coalescent method is less sensitive to different levels of missing data and ILS, a fact that has been highlighted by other studies (Mirarab & Warnow, 2015). This likely reflects its natural treatment of missing data, where missing taxa in each gene tree simply do not induce additional quartets.

Incongruences between nuclear and plastid phylogenies, commonly observed in plants, have frequently been interpreted as evidence of hybridization, in the phenomenon known as chloroplast capture, when the genome of one lineage becomes fixed in another after hybridization (Rieseberg & Soltis, 1991; Stull & al., 2023). When comparing the results of our study with those of Loeuille & al. (2015), which included chloroplast markers, our results indicate the potential for chloroplast capture as an interpretation of cytonuclear discord. However, the plastid trees from that study have insufficient resolution to allow for a more direct comparison and we did not assemble chloroplast sequences from our results due to the dense multiplexing experimental design used for target capture, which minimizes the amount of off-target reads. While limited variation in Asteraceae plastid DNA has created challenges for smaller datasets (e.g., in resolving phylogenetic relationships in *Aldama*; Loeuille & al., 2021), whole chloroplast data generated with methods such as genome-skimming may overcome such limitations and could be a helpful tool in verifying some of these potential chloroplast capture events.

Ancient polyploidization events, which are frequently interpreted as a result of hybridization (i.e., allopolyploidy; see Soltis & al., 2007; Barker & al., 2016), have been moderately well-investigated in Asteraceae, with the origin of the family being associated with a hexaploid ancestor (Barker & al., 2016) and with many subsequent polyploidy events identified in other parts of the tree (Zhang & al., 2021). However, the tribe Vernonieae has been mostly overlooked in these studies, and no ancient polyploidy events have been identified in this tribe so far. Nevertheless, the variation in chromosome numbers in the tribe (n = 7 to n = 68; Keeley & Robinson, 2009; Angulo & al., 2022), especially in the New World species, could indicate repeated events of ancient hybridization and polyploidy impacting the diversification of the group. The reconstruction of chromosome numbers in Angulo & al. (2022) indicates a base number of n = 18 for the Lychnophorinae clade, although the sampling is limited and the topology differs from the one shown here. This previous reconstruction indicates an increase to n = 19 in the *Lychnocephalus* clade, decreases to n = 17 in *Minasia* and *Paralychnophora* and decreases to n = 17 and 15 in a clade mostly formed by *Lychnophora* and two *Eremanthus*. With this limited karyological data available for the group, it seems that dysploidy is a more impacting force than duplications in Lychnophorinae's evolutionary history. Expanding sampling of chromosome counts and dedicated studies using transcriptomic and genomic data are needed to clarify if polyploidy indeed played a role in this group's evolution.

Generic circumscription of Lychnophorinae. — The overall relationships within the subtribe are moderately similar to those found in previous molecular phylogenetic studies, either with phylogenomic data or not (Loeuille & al., 2015; Siniscalchi & al., 2019). One major change from these previous studies is the position of the monotypic genus *Allocephalus* Bringel & al. (currently included in subtribe Dipterocypselinae), which emerges as sister to all American Vernonieae in the ASTRAL analyses, while it emerges as sister to all Lychnophorinae in the concatenated analysis, as seen in Siniscalchi & al. (2019). Its position at the subtribal level is unclear due to our narrowed sampling focus on Lychnophorinae, but our results indicate that a close relationship with Lychnophorinae is uncertain. The genus displays an array of plesiomorphic characters of Lychnophorinae that are found in other lineages of Vernonieae (e.g., herbaceous habit, T-shaped trichomes, heads in dense glomerules) (Siniscalchi & al., 2019).

Cololobus H.Rob. (Fig. 1A) is included for the first time in a phylogeny and emerges as one of the early-divergent lineages of Lychnophorinae. This is a small genus of five shrubby species from eastern Brazil, which show type A pollen, glabrous corollas, style bases with a node, and persistent

phyllaries (Robinson, 1994; Monge & al., 2018). These features were used to place *Cololobus* in the non-monophyletic subtribe Vernoniinae (Robinson, 1994; Keeley & Robinson, 2009), but it lacks the typical resin ducts in the corolla lobes that most members of this subtribe have. *Cololobus* shares the thick indumentum covering stems and leaves, semi-amplexicaul leaf sheath, and fimbrillate receptacle with early-diverging Lychnophorinae (Loeuille & al., 2019). Regarding these unusual features of *Cololobus*, Robinson (1994) emphasized the glabrous corolla with short lobes when establishing the genus, but a similar corolla is found in several Lychnophorinae, e.g., *Albertinia* Spreng. and *Gorceixia* Baker. Similarly, the style with basal node is uncommon in Lychnophorinae but found in *Albertinia* (Loeuille & al., 2019). This evidence shows that *Cololobus* is morphologically better placed within Lychnophorinae than Vernoniinae, especially given the non-monophyletic aspect of the latter (Siniscalchi & al., 2019).

Relationships between genera. — The identification of numerous past reticulation events and subsequent high level of nuclear incongruence limit our capacity to reconstruct the phylogenetic relationships between several genera of Lychnophorinae and offer a mechanistic explanation for the low statistical support recovered in previous Lychnophorinae phylogenetic analyses (Loeuille & al., 2015). In the early-diverging lineages, reticulation events and a high level of nuclear incongruence make it difficult to interpret the non-monophyly of *Anteremanthus* H.Rob. and *Maschalostachys* (Fig. 1B), two small genera with well-defined morphological characteristics (Loeuille & Roque, 2017; Loeuille & al., 2019), especially *A. hatschbachii*, which shows near-perfect topological conflict (Fig. 3). In the concatenated analyses, this species emerges in the same clade as *A. piranii* together with *Hololepis* DC. and *Eremanthus crotonoides*. The sampling of members of these early-diverging lineages in phylogenies is still limited, and further addition of other early-diverging genera, such as *Piptocoma* Cass. (Aguilar-Cano & al., 2025) may bring some clarity regarding relationships in this part of the tree. The genus *Minasia* emerges as monophyletic in all analyses, on a fairly long branch with only minor conflict among two of its species and it is not involved in any of the possible hybridization events we identified.

The *Prestelia* and allies clade (*Eremanthus crotonoides*, *Hololepis*, *Lychnocephalus*, *Prestelia* Sch.Bip.), as defined in Loeuille & al., 2015, emerges as polyphyletic. *Prestelia* and *Lychnocephalus* (Fig. 1C,D) emerge in a more nested position than *Hololepis* and *E. crotonoides*. In our results, this *Lychnocephalus* + *Prestelia* clade (LPP 0.71, Fig. 3B) also includes three anomalous species of *Lychnophora*. *Lychnophora souzae* H.Rob. forms a clade with *L. brunioides* Mart., a relationship that had been previously proposed based on morphological affinities (e.g., obtriangular leaf sheath), and both were placed in the Incertae Sedis section as "*Lychnophora brunioides*" group of Loeuille & al. (2019). *Prestelia* and *Lychnocephalus* have semi-amplexicaul leaf sheath, which is also displayed by *Lychnophora albertinioides* Gardner (Fig. 1E), positioned here as sister to the three species of *Lychnocephalus*. However, *Lychnophora albertinioides* shows inflorescences in pseudoglomerules while members of *Lychnocephalus* present third-order syncephalia. One reticulation event is identified between *Lychnophora albertinioides* and *Lychnocephalus jolyanus* and the low inheritance probability (0.2) in the sister edge indicates it is likely an introgression. A dedicated study of this clade, including an expanded sampling is needed to clarify the morphological evolution of leaf and inflorescence traits. The position of *Proteopsis* remains unclear, as in previous phylogenetic studies (Loeuille & al., 2015). This taxon produces a high level of discordant topologies between the main clades and is also involved in one recent reticulation event, which could indicate introgression with *Chronopappus bifrons*. *Proteopsis* shares the uncommon pseudoglomerule inflorescence with *Piptolepis* but is unique by the combination of a caulirosulate habit, amplexicaul leaf sheath and phyllaries with spiny appendages (Loeuille & al., 2019). The inclusion of the second species of the genus in a future phylogenomic study would be helpful to ascertain the relationships of *Proteopsis* within Lychnophorinae.

In the derived Lychnophorinae, *Lychnophorella* emerges as sister to *Eremanthus* with strong support, and *Lychnophora* as sister to the *Lychnophorella* + *Eremanthus* clade, with low support (Fig. 3B). A similar relationship was found by Siniscalchi & al. (2019), except that in some analyses, *Piptolepis* was sister to *Eremanthus* + *Lychnophorella*, but not by Loeuille & al. (2015) where the *Lychnophorella* and *Piptolepis* clades successively emerge as sister to a *Lychnophora* + *Eremanthus* clade. The lower support for the relationship between *Lychnophora* and the clade containing *Lychnophorella* + *Eremanthus* is potentially due to past reticulation events between *Eremanthus* and *Lychnophora* or *Piptolepis*, explored in the next paragraphs below. This new topology leads to a more complex scenario for the evolution of the leaf sheath; *Eremanthus* and *Lychnophora* are the only two genera without leaf sheaths in the whole subtribe, and *Lychnophorella* has pad-like leaf sheaths (Lusa & al., 2018; Loeuille & al., 2019). As this pad-like leaf sheath is also found in *Piptolepis*, its absence in *Eremanthus* and *Lychnophora* is probably secondary, but we cannot exclude the possibility of two independent losses or an independent acquisition of pad-like leaf sheath in *Lychnophorella* and *Piptolepis*. Anatomical studies are currently underway to better understand the evolution of leaf sheath in Lychnophorinae. In the present study, *Piptolepis* is more closely related to the *Lychnocephalus* + *Prestelia* clade as discussed above than the *Lychnophora*, *Eremanthus*, and *Lychnophorella* clades, differing from the topologies of Loeuille & al. (2015) (no *Lychnocephalus* representative was included in Siniscalchi & al., 2019). While *Piptolepis* shares few morphological traits with *Prestelia* or *Lychnocephalus*, some authors (Coile & Jones, 1981; Semir, 1991) have suggested placing the three anomalous species of *Lychnophora* (*L. albertinioides*, *L. brunioides*, *L. souzae*) that emerge in the *Lychnocephalus*

+ *Prestelia* clade in our study in *Piptolepis*. *Lychnophora albertinioides* has similar leaf shape, indument and pseudoglomerules to *Piptolepis oleaster* Sch.Bip. (Loeuille & al., 2019). The similarity of *L. brunioides* and *L. souzae* to *Piptolepis* is based on the interpretation of the obtriangular leaf sheath as a variant of the pad-like leaf sheath. Additionally, some *Piptolepis* species have pedunculate heads like *L. souzae* but the syncephalium of *L. brunioides* is not found in *Piptolepis* (Loeuille & al., 2019). Interestingly, our analyses indicate that one reticulation event happened between *Prestelia* and the stem of the *Piptolepis* clade, indicating a potential hybrid origin of *Prestelia*.

Monophyly and relationships within *Piptolepis*. — *Piptolepis* is monophyletic if *P. pabstii* (Fig. 1G) and *P. rosmarinifolia* (Fig. 5F) are excluded as they emerge as a subclade within *Eremanthus* (Fig. 2). These two species are endemic to the *campos rupestres* of the Goiás Plateau occurring in disjunction from the other *Piptolepis* species in Minas Gerais (Loeuille, 2024). *Piptolepis pabstii* and *P. rosmarinifolia* are morphologically similar in sharing a subshrubby, caespitose habit unbranched or sparsely branched towards the apex, with xylopodium, narrow linear leaves and capitula fused in a syncephalium with only one floret per capitulum (Bringel & al., 2019). The only feature these two species share with the rest of *Piptolepis* is the presence of a pad-like leaf sheath (Bonfim e Cândido & al., 2025). The characters used by Loeuille & al. (2019) to define *Piptolepis* and to support the inclusion of *P. rosmarinifolia* and *P. pabstii* (3- to 5-armed swollen trichomes and biseriate pappus) are not present in all species of the genus and occur in other genera of Lychnophorinae. Thus, we propose excluding these two species from *Piptolepis* to maintain monophyly of this genus, to be addressed in a future work.

Numerous inferred reticulation events including extant lineages, as well as topological conflicts and low branch support make the resolution in that clade unstable, but some

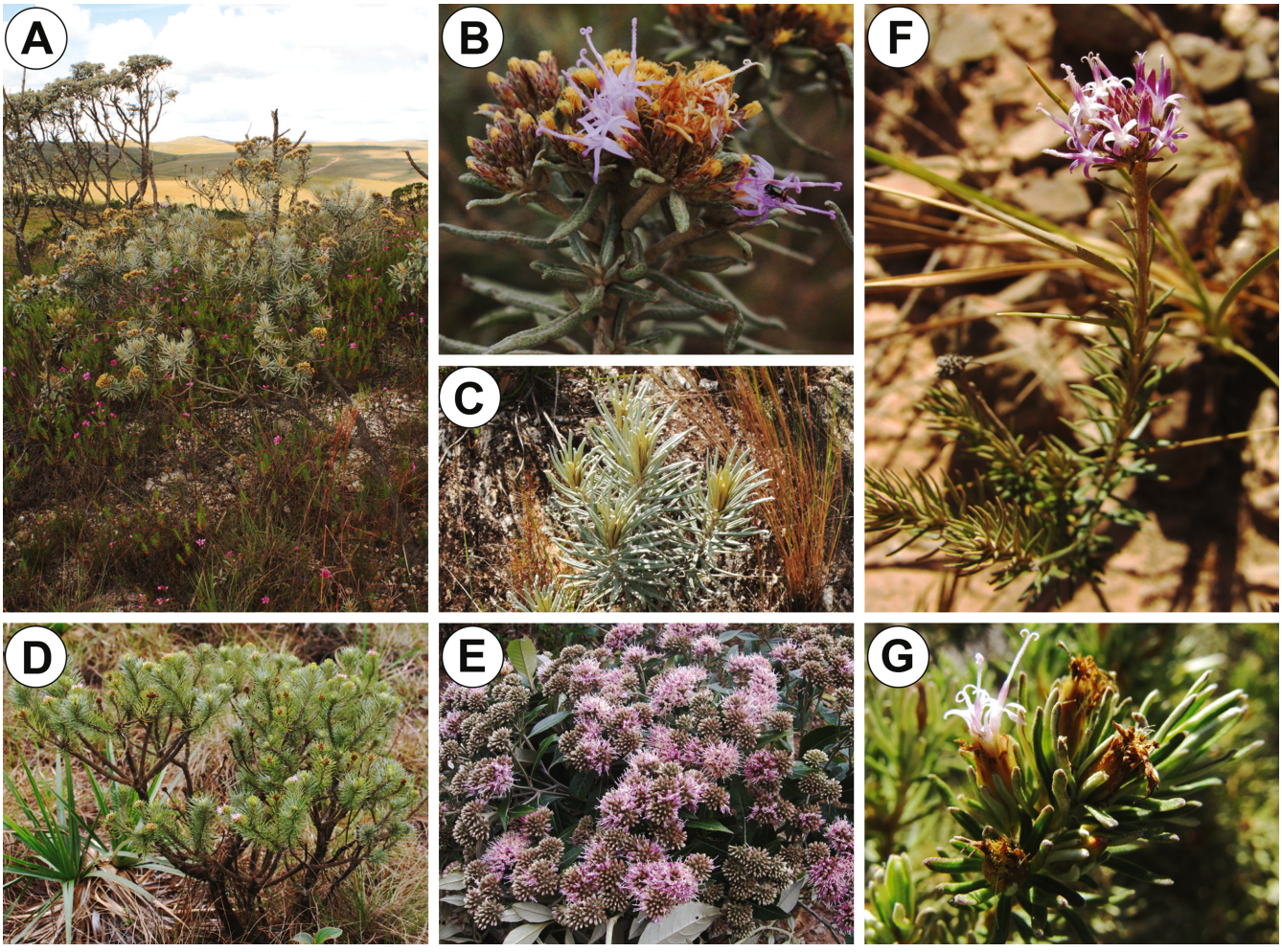

Fig. 5. Hybrid lineages and potential sister taxa in Lychnophorinae. **A**, *Lychnophora penninervia*, habit (note co-occurring *Eremanthus erythropappus* in the background on the left); **B**, *Lychnophora penninervia*, secondary heads; **C**, *Lychnophora penninervia*, branch tips (note strongly revoluted leaves); **D**, *Lychnophora ericoides*, habit; **E**, *Eremanthus erythropappus*, flowering secondary heads (note leaves with wide, expanded blades on the bottom); **F**, *Piptolepis rosmarinifolia*, habit and secondary head; **G**, *Piptolepis ericoides*, branch tip showing simple heads. — Photo credits: A–C by C.M. Siniscalchi; D–E by B. Loeuille; F & G by J.B. Cândido.

relationships are found in all analyses. *Piptolepis oleaster* always emerges as the sister-group to the remaining species of the genus. A clade containing *P. corymbosa* J.B.Cândido & Loeuille, *P. pilosa* J.B.Cândido & Loeuille, *P. imbricata* Sch.Bip., *P. glaziouana* Beauverd, *P. leptospermoides* Sch. Bip. (Fig. 1F) and *P. riparia* Loeuille & al. is constantly recovered with good statistical support, but internally unresolved. This group is characterized by the presence of small leaves, ranging from 2–15 × 0.8–9 mm (Bonfim e Cândido & al., 2025). Additionally, there is evidence of a well-supported reticulation event involving ancestors of this clade (Fig. 4F). In some analyses, *P. procumbens* J.B.Cândido & Loeuille also emerges in that clade but in others it emerges as sister to the clade formed by *P. campestris* and *P. gardneri* Baker. Even though *P. procumbens* has the small leaves typical of that clade, it shares some features with *P. campestris* and *P. gardneri*: ligulate pad-like leaf sheath, fimbrillate receptacle, pseudoglomerule inflorescence, and prismatic cypselae. However, *P. campestris* and *P. gardneri* differ from *P. procumbens* by their treelet habit (vs. shrub), larger leaves (8–68 × 3–10 mm), and uniseriate pappus with subequal or equal setae (vs. biseriate pappus with unequal setae) (Marques & al., 2022; Bonfim e Cândido & al., 2025). Despite *P. redacta* having small leaves (3–10 × 1.5–3.6 mm), it emerges as sister to a clade composed of *P. pseudomyrtus*, *P. ericoides*, and *P. schultziana* Loeuille & D.J.N.Hind. These species co-occur in a small area in the Espinhaço Range in the state of Minas Gerais (Congonhas do Norte and Alvorada de Minas). They are similar by their ascending leaves with flat margins and hyphodromous venation (except *P. pseudomyrtus* with eucamptodromous venation), inflorescence in terminal raceme and pappus series subequal or equal in size (Bonfim e Cândido & al., 2025). *Piptolepis monticola* Loeuille emerges as the sister-group to the clade composed of *P. redacta*, *P. pseudomyrtus*, *P. ericoides*, and *P. schultziana* and morphologically resembles the aforementioned species in abaxial leaf surface indumentum (velutinous except for *P. ericoides* with canescent indumentum), campanulate involucre (except *P. schultziana* with cylindrical involucre), and pappus series subequal or equal in size.

Monophyly of *Eremanthus*. — In its current circumscription, *Eremanthus* is not monophyletic due to the position of *Piptolepis pabstii* and *P. rosmarinifolia* that are nested within it, and the emergence of two species of *Eremanthus, E. hatschbachii* H.Rob. and *E. veadeiroensis* H.Rob, within *Lychnophorella*. Ours is the first phylogeny to include *E. hatschbachii*, a species that is at odds within *Eremanthus* due to its pad-like leaf sheath, which was overlooked in previous studies (leaves are sheathless in *Eremanthus*) and its poorly branched shrubby habit (vs. mostly treelet habit in the genus). The species is morphologically related to *Lychnophorella leucodendron* (Mattf.) Loeuille & al. and *L. saxicola* J.B.Cândido & Loeuille; its congested capitula in a dichasium (Bonfim e Cândido & Loeuille, 2020) fits very well in *Lychnophorella*. In a previous phylogeny, *E. veadeiroensis* was recovered deeply nested in *Eremanthus*, in a subclade with *E. argenteus* MacLeish & H.Schumach. and *E. elaeagnus*, which are the morphologically closest species to it (Loeuille & al., 2015). Here, it is recovered inside *Lychnophorella* in all analyses with good support and no apparent conflicts. *Eremanthus veadeiroensis* lacks the typical pad-like leaf sheath of *Lychnophorella* and shares with *Eremanthus* the treelet habit with sheathless leaves with flat margins, indumentum composed of 3- to 5-armed swollen trichomes and terminal cymose pedunculate inflorescence (Loeuille & al., 2019). The position we found for *E. veadeiroensis* could indicate a spurious effect of the varying position of *E. elaeagnus* in different topologies, its morphologically closest species, with which it shares silvery leaves, heads in glomerule, pubescent cypselae, and straight pappus setae. These differences between our work and the relationships found by Loeuille & al. (2015) could indicate deep chloroplast vs. nuclear incongruence, as this previous work was based on ribosomal and plastid markers. Also, it could indicate an effect of taxonomic sampling, as only 7 out of 22 species of *Eremanthus* are included in the present study, and further sampling could clarify the position of *E. elaeagnus* and *E. veadeiroensis* in relation to other genera. Finally, *E. crotonoides* falls completely outside the current circumscription of the genus, emerging among the early-derived members of the tribe and confirming results from Loeuille & al. (2015). However, the lack of resolution in that part of the tree and the occurrence of several reticulation events preclude the transfer of this taxon to another genus at the moment.

Lychnophora penninervia Sch.Bip. (Fig. 5A–C) has been placed under the synonymy of *Eremanthus reticulatus* by previous authors (Baker, 1873; Coile & Jones, 1981; Semir, 1991; Loeuille & al., 2019). *Eremanthus reticulatus* was first described in *Lychnophora* but was excluded in the revision of the genus by Coile & Jones (1981). According to Loeuille & al. (2019), it belongs to a small species complex with *E. syncephalus* (Sch.Bip.) Loeuille & al. and an unpublished putative new species. There are few specimens available for these taxa and the species circumscriptions are still unclear. Loeuille & al. (2019) placed them within *Eremanthus* based on the position of *E. syncephalus* in phylogenetic studies (Loeuille & al., 2015) and a combination of morphological features: flat leaf margins, syncephalia lacking reduced leaves between the capitula and turbinate cypsela. However, the type material of *L. penninervia* (Serra de Ouro Branco, *Riedel 2952*, P02538890) and some recent collections (Serra da Canastra, *Siniscalchi 443*, HUFU73332; Parque Nacional das Sempre Vivas, *Chaves 102*, ALCB138858; Diamantina, *Cândido 295*, UFP88612) show leaves with revolute margins that are not found neither in *E. reticulatus*, nor in the genus *Eremanthus* (Loeuille & al., 2019). We therefore consider *L. penninervia* as a distinct species, included in our analyses with the specimen *Cândido 295* (Appendix 1). All the species of this complex (*L. penninervia*, *E. reticulatus*, *E. syncephalus*) display a mixture of morphological features of *Eremanthus* and *Lychnophora*, already noted by Gardner

(1846) in the description of *E. reticulatus*. Our results indicate that *L. penninervia* emerges as sister to *Lychnophora* in the analysis carried out with the total dataset and within *Eremanthus* in the analyses with the 68p dataset (suppl. Figs. S1–S9), which potentially indicates a role of missing data on positioning this species on the trees. It is also a part of a twice-recovered reticulation event with *Lychnophora ericoides* Mart. (Fig. 5D), with high bootstrap support (Fig. 4C,G). Field observations indicate that only single individuals have been found in localities with large populations of *L. ericoides* and *E. erythropappus* (Fig. 5E), which are common species that form large populations with overlapping flowering periods. This could lead to reproductive barriers being broken by large quantities of pollen produced by both parental species, giving rise to homoploid hybrids, as both potential parental species have $x = 17$ (Mansanares & al., 2001, 2002; Salles-de-Melo & al., 2010). Despite both parental species being widespread in *campos rupestres* of Minas Gerais State, only a handful of *L. penninervia* specimens have been collected, a similar pattern to the one found for two other intergeneric hybrids in Lychnophorinae (Loeuille & al., 2015). Here, we consider *Lychnophora* ×*penninervia* a natural hybrid between *Lychnophora ericoides* and *Eremanthus erythropappus*. More studies are needed to understand if the other species of the *E. reticulatus* complex are hybrids as well.

Eremanthus plays an intriguing role in the complex evolutionary history of Lychnophorinae. Loeuille & al. (2015) proposed that most of the diversification of the subtribe occurred in the *campos rupestres*, with a few species expanding into the *cerrado*, but without significant diversification in this savanna biome. *Eremanthus* has the widest geographic distribution among all Lychnophorinae and contains a few widespread species that occur in many biomes that surround *campos rupestres*, such as *E. erythropappus, E. capitatus*, and *E. glomerulatus* in Atlantic Forest and Cerrado. This wide distribution could have led to many opportunities for contact with *campos rupestres*-restricted species during favorable climatic conditions (Anderson & Stebbins, 1954; Folk & al., 2018b); changing geographic ranges are thought to be critical for hybridization opportunities and establishment. Some of these widespread species form large populations that flower abundantly, which associated to a broad guild of pollinators attracted to their non-specialized flowers may have led to a wider area for pollen dispersal (Loeuille & al., 2019). The fact that *Eremanthus* contains several identified hybrid edges and species in disparate clades suggests a history of flickering connectivity potentialized by the changing populational dynamics of these species.

Hybridization in connection with biogeography and diversification. — This extensive history of reticulation can be understood in light of the biogeographical history of Lychnophorinae and the timing of its diversification. Although we do not present new analyses in this respect here, Vasconcelos & al. (2020) dated and analyzed the diversification rates of the tree from Loeuille & al. (2015). Their analysis (available in their Supplementary Materials) shows a diversification regime shift to higher diversification rates comprehending almost the whole subtribe, and a crown age younger than 2 Ma for most of the more nested clades. Similar but slightly older ages were found by Mandel & al. (2019, their suppl. fig. S5), from which we re-used some sequences in this study. Lychnophorinae are highly endemic to *campos rupestres*, especially those found on the Espinhaço Range, with few taxa occurring in *cerrado* and tropical seasonal forests (Loeuille & al., 2015, 2019). While the role of Pleistocene climate change and the formation of refugia on the diversification of *campos rupestres* plant lineages has been debated (e.g., Vasconcelos & al., 2020; Rapini & al., 2021), the extent of hybridization and high diversification rates of Lychnophorinae align well with models of flickering connectivity (Flantua & al., 2019), initially proposed to explain the diversity of another Neotropical montane biome, the *paramos*. In the flickering connectivity model, repeated cycles of contact and isolation of populations would vary spatially and temporally creating multiple opportunities for contact between populations of different species that would then cross and become isolated again allowing for hybrid speciation (reviewed in Anderson & Stebbins, 1954; Folk & al., 2018a; Schley & al., 2022). The role of multiple cycles of contact and separation during the Quaternary on the diversification of components of the *campos rupestres* flora is a long-standing hypothesis (e.g. Giulietti & Pirani, 1988; Giulietti & al., 1997), with which our results are consistent. Explicit habitat models through time (e.g., Folk & al., 2018b, 2023) are needed to further test whether habitat connectivity directly predicts hybridization.

The repeated appearance of particular habits (e.g., caulirosulate), types of syncephalia, and other traits in species of Lychnophorinae could reflect the process called "combinatorial speciation", in which variants accumulated in divergent lineages are passed around through hybridization (Marques & al., 2018), thus generating apparent convergent features in multiple lineages. Trichome complement has been used to track putative hybrids in *Vernonia* (Hunter & Austin, 1967; Faust & Jones, 1973). Wagner & al. (2014) studied the trichomes of Lychnophorinae and found results considered as independent acquisition within the current phylogenetic hypothesis: the shared presence of unbranched auriculate trichomes (type A3) in *Proteopsis* and *Piptolepis* spp., of T-shaped trichomes with arms diagonal to the stalk (type C3) in *Lychnocephalus humillimus* (Sch.Bip.) Loeuille & al. and *Prestelia espeletoidea* Loeuille & al., and of stellate bladder-like trichomes (type E2) in *Eremanthus erythropappus*, *Lychnophorella santosii* and *Piptolepis oleaster*. In each case, these lineages were involved in hybridization events (Fig. 4), suggesting that the presence of these trichomes may have been inherited through hybridization. With its high diversity and morphological specialization restricted to isolated and climatically challenged environments, Lychnophorinae could be the counterpart of another iconic montane Asteraceae group, the subtribe Espeletiinae of tribe Millerieae

(commonly known as *frailejones*). Recent studies on Espeletiinae have shown a similar scenario: rapid diversification in the Pleistocene, with high levels of hybridization among sympatric species (Pouchon & al., 2021). While the role of the Andes rising process and its steep elevational incline cannot be compared to the stable geological relief of the *campos rupestres*, the edaphic specialization shown by many species of Lychnophorinae can effectively restrict the range in which these species can expand, simulating the effect of altitudinal ranges and their climatic limitations.

Future perspectives. — While our study advances our knowledge of the history of Lychnophorinae, uncovering multiple events of ancient hybridization that explain conflicts found in this and previous phylogenies, it still leaves unanswered questions. Improved sampling in some clades, such as *Eremanthus* and the early-divergent lineages, could bring clarity for some of the relations that change among topologies. *Eremanthus*, in particular, needs a densely sampled phylogeny, including multiple accessions of all species and the identified natural hybrids, as this genus seems to be the source and destination of this complicated history of ancient hybridization. A population-level sampling in certain key groups, such as the *Eremanthus reticulatus/Lychnophora ×penninervia* complex, can help unveil the morphological variation and delineate species boundaries, while confirming the role of hybridization in its formation. Our results support some future taxonomic changes, such as the inclusion of *Eremanthus hatschbachii* in *Lychnophorella* and the exclusion of *Piptolepis pabstii* and *P. rosmarinifolia* from *Piptolepis*, but some long-standing issues, such as the position of *Lychnophora brunioides* as "incertae sedis", still remain. Finally, the use of a different, newly developed probe set, the Compositae-ParaLoss-1272 (Moore-Pollard & al., 2024), which contains longer orthologous loci and improves issues of paralogy and incongruence seen with the Compositae1061 set, could generate a more cohesive dataset that could refine the resolution of deeply nested relationships in Lychnophorinae.

Our study indicates that potential hybridization can be more thoroughly investigated in phylogenomic studies that present low support and high incongruence, common features of phylogenetic studies in Asteraceae, as it can help disentangle ILS from hybridization. Our results also add to a growing corpus of evidence on the role of hybridization in plant evolution and the challenges it poses to taxonomic resolutions. With the increasing availability of large phylogenomic datasets and their use as evidence for new classifications or to split and merge genera and species, empirical data like ours could be an important source in developing future models that can capture the complexities of genetic exchange across species boundaries.

■ DATA AVAILABILITY

Raw sequence data is available at the NCBI Sequencing Read Archive, Bio Projects PRJNA1108325, PRJNA546287 and PRJNA540287. All supplementary figures, tables, data files and scripts are available at Zenodo (https://zenodo.org/records/15426471) and GitHub (https://github.com/carol-siniscalchi/lychnophorinae-reticulation).

■ AUTHOR CONTRIBUTIONS

JBC, CMS and BL conceived this study, with input from all authors. JBC and BL collected samples from the field and herbaria; ERMP, JLG and SD generated the new molecular data; CMS and JRM analyzed the data; JRM, RAF and JRP contributed to data interpretation; CMS, BL and JBC wrote the initial draft of this manuscript. All authors read, commented on, and approved the submitted version.

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Appendix 1. Vouchers used in this study.

Information is presented in the following order: species name with authority, country, state, collector name and number, herbarium catalog number/barcode, and NCBI Sequence Read Archive run number. * indicates new sequences obtained in this study.

Albertinia brasiliensis Spreng., Brazil: Bahia, *Siniscalchi, C.M. 466* (SPF00222754) [SRR9119087]. ***Allocephalus gamolepis*** J.Bringel, J.N.Nakaj. & H.Rob., Brazil: Goiás, *Bringel, J. 416* (CEN00075683) [SRR9119088]. ***Anteremanthus hatschbachii*** H.Rob., Brazil: Minas Gerais, *Loeuille, B. 443* (SPF00192183) [SRR28925559]*. ***Anteremanthus piranii*** Roque & F.A.Santana, Brazil: Bahia, *Roque, N. &. al. 4408* (ALCB115483) [SRR28925560]*. ***Blanchetia heterotricha*** DC., Brazil: Bahia, *Loeuille, B. 680* (SPF00215040) [SRR28925545]*. ***Centratherum punctatum*** Cass., Brazil: Goiás, *Siniscalchi, C.M. 498* (SPF00222676) [SRR9119034]. ***Chronopappus bifrons*** (DC. ex Pers.) DC., Brazil: Minas Gerais, *Loeuille, B. 465* (SPF00189308) [SRR9119076]. ***Cololobus rupestris*** (Gardner) H.Rob., Brazil: Minas Gerais, *Loeuille, B. 886* (SPF00216078) [SRR28925557]*. ***Eremanthus auriculatus*** MacLeish & H.Schumach., Brazil: Goiás, *Loeuille, B. 845* (SPF00216010) [SRR9208068]. ***Eremanthus capitatus*** (Spreng.) MacLeish, Brazil: Goiás, *Loeuille, B. 850* (SPF00216015)

Appendix 1. Continued.

[SRR28925528]*. ***Eremanthus crotonoides*** (DC.) Sch.Bip., Brazil: Espírito Santo, *Fraga 2347* (MBM392634) [SRR9208069]. ***Eremanthus elaeagnus*** (Mart. ex DC.) Sch.Bip., Brazil: Minas Gerais, *Cândido, J. 389* (BHCB208897) [SRR28925519]*. ***Eremanthus erythropappus*** (DC.) MacLeish, Brazil: Minas Gerais, *Loeuille, B. 861* (SPF00216061) [SRR9208070]. ***Eremanthus hatschbachii*** H.Rob., Brazil: Bahia, *Cândido, J. 279* (UFP88474) [SRR28925518]*. ***Eremanthus incanus*** (Less.) Less., Brazil: Bahia, *Siniscalchi, C.M. 627* (SPF00224451) [SRR9118985]. ***Eremanthus reticulatus*** (Gardner) Loeuille, Semir & Pirani, Brazil: Minas Gerais, *Cândido, J. 295* (UFP88612) [SRR28925543]*. ***Eremanthus uniflorus*** MacLeish & H.Schumach., Brazil: Minas Gerais, *Loeuille, B. 914* (SPF00221423) [SRR28925517]*. ***Eremanthus veadeiroensis*** H.Rob., Brazil: Goiás, *Loeuille, B. 847* (SPF00216012) [SRR28925516]*. ***Gorceixia decurrens*** Baker, Brazil: Minas Gerais, *Loeuille, B. 821* (SPF00216020) [SRR9119177]. ***Heterocoma ekmaniana*** (Philipson) Loeuille, J.N.Nakaj. & Semir, Brazil: Goiás, *Siniscalchi, C.M. 494* (SPF00222675) [SRR9119027]. ***Heterocoma lanuginosa*** (Glaz. ex Oliv.) Loeuille, J.N.Nakaj. & Semir., Brazil: Minas Gerais, *Loeuille, B. 635* (SPF00214883) [SRR28925515]*. ***Hololepis pedunculata*** (DC. ex Pers.) DC., Brazil: Minas Gerais, *Siniscalchi, C.M. 588* (SPF00224422) [SRR9119023]. ***Lychnocephalus cipoensis*** Semir ex F.M.B.Gomes & Loeuille, Brazil: Minas Gerais, *Cândido, J. 361* (UFP87596) [SRR28925561]*. ***Lychnocephalus jolyanus*** Semir ex F.M.B.Gomes & Loeuille, Brazil: Minas Gerais, *Gomes, F.M.B. 7* (UFP88328) [SRR28925536]*. ***Lychnocephalus tomentosus*** Mart. ex DC., Brazil: Minas Gerais, *Cândido, J. 288* (UFP87600) [SRR28925537]*. ***Lychnophora albertinioides*** Gardner, Brazil: Minas Gerais, *Cândido, J. 368* (ALCB073853) [SRR28925538]*. ***Lychnophora brunioides*** Mart., Brazil: Minas Gerais, *Loeuille, B. 466* (UEC015100) [SRR9208062]. ***Lychnophora candelabrum*** Sch.Bip., Brazil: Minas Gerais, *Cândido, J. 310* (BHCB211943) [SRR28925539]*. ***Lychnophora ericoides*** Mart., Brazil: Minas Gerais, *Cândido, J. 302* (UFP88607) [SRR28925540]*. ***Lychnophora granmogolensis*** (Duarte) Semir, Brazil: Bahia, *Siniscalchi, C.M. 348* (SPF00222746) [SRR9119165]. ***Lychnophora haplopappa*** Loeuille, Semir & Pirani, Brazil: Minas Gerais, *Loeuille, B. 902* (SPF00216064) [SRR9208063]. ***Lychnophora hatschbachii*** (H.Rob.) Loeuille, Semir & Pirani, Brazil: Minas Gerais, *Loeuille, B. 613* (SPF00214898) [SRR9208073]. ***Lychnophora itacambirensis*** Semir & Leitão Filho, Brazil: Minas Gerais, *Siniscalchi, C.M. 528* (SAMES13391) [SRR28925541]*. ***Lychnophora passerina*** (Mart. ex DC.) Gardner, Brazil: Minas Gerais, *Cândido, J. 335* (BHCB211939) [SRR28925542]*. ***Lychnophora salicifolia*** Mart., Brazil: Minas Gerais, *Cândido, J. 264* (UFP88438) [SRR28925544]*. ***Lychnophora souzae*** H.Rob., Brazil: Minas Gerais, *Cândido, J.B. 308* (HUFU00080080) [SRR28925547]*. ***Lychnophora uniflora*** Sch.Bip., Brazil: Bahia, *Cândido, J.B. 266* (UFP88440) [SRR28925548]*. ***Lychnophorella bishopii*** (H.Rob.) Loeuille, Semir & Pirani, Brazil: Bahia, *Cândido, J.B. 278* (HUEFS260205) [SRR28925549]*. ***Lychnophorella leucodendron*** Loeuille, Semir & Pirani, Brazil: Bahia, *Loeuille, B. 670* (SPF00215023) [SRR9208075]. ***Lychnophorella morii*** (H.Rob.) Loeuille, Semir & Pirani, Brazil: Bahia, *Loeuille, B. 658* (SPF00215033) [SRR9208072]. ***Lychnophorella santosii*** (H.Rob.) Loeuille, Semir & Pirani, Brazil: Bahia, *Siniscalchi, C.M. 354* (SPF00222660) [SRR36950179]. ***Maschalostachys markgrafii*** (G.M.Barroso) Loeuille & Roque, Brazil: Minas Gerais, *Cândido, J.B. 383* (UFP88570) [SRR28925550]*. ***Maschalostachys mellosilvae*** Loeuille & Roque, Brazil: Minas Gerais, *Loeuille, B. 827* (SPF00216021) [SRR9208074]. ***Minasia alpestris*** (Gardner) H.Rob., Brazil: Minas Gerais, *Cândido, J.B. 286* (UFP88567) [SRR28925551]*. ***Minasia pereirae*** H.Rob., Brazil: Minas Gerais, *Loeuille, B. 862* (SPF00216032) [SRR9119138]. ***Minasia scapigera*** H.Rob., Brazil: Minas Gerais, *Cândido, J.B. 284* (UFP88571) [SRR28925552]*. ***Minasia splettiae*** H.Rob., Brazil: Minas Gerais, *Siniscalchi, C.M. 562* (SPF00224403) [SRR28925553]*. ***Paralychnophora atkinsiae*** D.J.N.Hind, Brazil: Bahia, *Cândido, J.B. 276* (UFP88437) [SRR28925554]*. ***Paralychnophora glaziouana*** Loeuille, Brazil: Minas Gerais, *Cândido, J.B. 289* (UFP88613) [SRR28925555]*. ***Paralychnophora harleyi*** (H.Rob.) D.J.N.Hind, Brazil: Bahia, *Ferreira, S.C. 265* (HUEFS142763) [SRR9119184]. ***Piptocarpha macropoda*** (DC.) Baker., Brazil: Minas Gerais, *Forzza, R. 4276* (NY1039146) [SRR28925556]*. ***Piptolepis redacta*** J.B.Cândido & Loeuille, Brazil: Minas Gerais, *Cândido, J.B. 353* (UFP88701) [SRR28925525]*. ***Piptolepis campestris*** Semir & Loeuille, Brazil: Minas Gerais, *Cândido, J.B. 299* (UFP88664) [SRR28925558]*. ***Piptolepis corymbosa*** J.B.Cândido & Loeuille, Brazil: Minas Gerais, *Cândido, J.B. 331* (HUFU00080172) [SRR28925535]*. ***Piptolepis ericoides*** Sch.Bip., Brazil: Minas Gerais, *Loeuille, B. 866* (SPF00216050) [SRR9119179]. ***Piptolepis gardneri*** Baker, Brazil: Minas Gerais, *Cândido, J.B. 336* (HUFU00080089) [SRR28925514]*. ***Piptolepis glaziouana*** Beauverd, Brazil: Minas Gerais, *Cândido, J.B. 323* (UFP88709) [SRR28925513]*. ***Piptolepis imbricata*** (Gardner) Sch.Bip., Brazil: Minas Gerais, *Cândido, J.B. 365* (UFP88667) [SRR28925534]*. ***Piptolepis leptospermoides*** (Mart. ex DC.) Sch.Bip., Brazil: Minas Gerais, *Cândido, J.B. 304* (HUFU00080084) [SRR28925533]*. ***Piptolepis monticola*** Loeuille, Brazil: Minas Gerais, *Cândido, J.B. 342* (UFP88685) [SRR28925532]*. ***Piptolepis oleaster*** (Mart. ex DC.) Sch.Bip., Brazil: Minas Gerais, *Cândido, J.B. 328* (UFP88665) [SRR28925531]*. ***Piptolepis pabstii*** (G.M.Barroso) Loeuille, Semir & Pirani, Brazil: Goiás, *Cândido, J.B. 250* (UFP88688) [SRR28925530]*. ***Piptolepis pilosa*** J.B.Cândido & Loeuille, Brazil: Minas Gerais, *Cândido, J.B. 376* (HUFU00080170) [SRR28925529]*. ***Piptolepis procumbens*** J.B.Cândido & Loeuille, Brazil: Minas Gerais, *Cândido, J.B. 324* (UFP88707) [SRR28925527]*. ***Piptolepis pseudomyrtus*** (A.St.-Hil.) Sch.Bip., Brazil: Minas Gerais, *Cândido, J.B. 345b* (UFP88696) [SRR28925526]*. ***Piptolepis riparia*** Loeuille, Semir & Pirani, Brazil: Minas Gerais, *Cândido, J.B. 382* (UFP88690) [SRR28925524]*. ***Piptolepis rosmarinifolia*** Bringel, J.B.Cândido & Loeuille, Brazil: Goiás, *Cândido, J.B. 251* (UFG 72843) [SRR28925546]*. ***Piptolepis schultziana*** Loeuille & D.J.N.Hind, Brazil: Minas Gerais, *Cândido, J.B. 354* (UFP88672) [SRR28925523]*. ***Prestelia eriopus*** Sch.Bip., Brazil: Minas Gerais, *Pirani, J.R. 6286* (SPF00222070) [SRR28925522]*. ***Proteopsis argentea*** Mart. & Zucc. ex Sch.Bip., Brazil: Minas Gerais, *Loeuille, B. 444* (SPF00192221) [SRR28925521]*. ***Pseudostifftia kingii*** H.Rob., Brazil: Bahia, *Roque N. 4490* (US1256220) [SRR9119050]. ***Vernonanthura polyanthes*** (Sprengel) Vega & Dematteis, Brazil: Minas Gerais, *Loeuille, B. 493* (K001066171) [SRR28925520]*.